

HAMEDAN, Iran, Islamic Republic ofWMO: **407680**Lat: **34.8688N**Lon: **48.5344E**Elev: **5738**StdP: **11.89**Time Zone: **3.50 (IRN)**Period: **96-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.2	8.7	-2.7	5.9	10.3	2.5	7.7	20.0	23.6	44.5	21.4	41.8	0.8	120	0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.9	97.2	60.9	95.2	60.5	93.2	59.8	65.3	89.9	63.1	88.7	61.5	88.1	8.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
55.7	81.8	78.1	52.9	73.9	75.9	50.5	67.5	72.4	33.5	90.0	31.6	88.5	30.2	88.0	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.5	18.6	15.8	DB	-2.5	100.7	7.4	1.7	-7.8	101.9	-12.2	102.8	-16.4	103.8	-21.8	105.0	(4)
(5)			WB	-2.7	66.9	7.6	3.3	-8.1	69.3	-12.5	71.2	-16.8	73.1	-22.3	75.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.1	29.0	34.2	44.4	52.4	60.9	70.9	77.4	76.2	67.3	56.7	43.4	35.1	(6)	
	DBStd	17.38	10.63	8.89	7.21	5.90	4.68	4.19	3.52	3.13	4.08	5.41	6.53	7.67	(7)	
	HDD50	2021	650	443	202	41	0	0	0	0	0	10	213	462	(8)	
	HDD65	5001	1115	862	639	380	142	6	0	0	23	259	649	926	(9)	
	CDD50	3522	0	1	28	112	338	626	850	813	520	219	14	1	(10)	
	CDD65	1026	0	0	0	1	15	182	385	348	92	3	0	0	(11)	
	CDH74	16771	0	0	1	29	482	2979	5631	5306	2146	197	0	0	(12)	
	CDH80	8464	0	0	0	2	78	1324	3219	2990	839	12	0	0	(13)	
	WSAvg	5.1	4.1	5.3	7.2	6.9	5.9	5.2	5.6	5.0	4.6	4.5	3.6	3.5	(14)	
	PrecAvg	13.1	1.4	1.5	2.2	2.2	1.1	0.2	0.2	0.0	0.1	0.9	1.8	1.6	(15)	
(16) Precipitation	PrecMax	20.2	2.4	3.2	4.2	4.2	2.8	0.8	0.9	0.4	1.5	3.2	8.4	5.2	(16)	
	PrecMin	10.1	0.4	0.2	0.5	0.4	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.0	(17)	
	PrecStd	2.3	0.6	0.7	0.9	1.0	0.7	0.2	0.2	0.1	0.3	1.0	1.6	1.2	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.3	60.2	72.2	77.5	85.6	96.4	100.7	98.8	92.3	81.3	67.9	60.8	(19)	
		MCWB	39.9	43.4	47.5	53.0	56.9	60.0	61.6	61.5	56.9	52.6	47.0	43.0	(20)	
	2%	DB	51.6	55.2	67.7	72.9	82.2	93.1	98.4	96.9	89.9	78.9	63.4	55.4	(21)	
		MCWB	38.9	41.4	45.6	51.3	55.3	59.9	61.2	61.1	56.4	52.2	45.9	41.9	(22)	
	5%	DB	47.8	51.7	62.9	69.5	79.1	90.0	96.4	95.0	87.8	76.2	59.5	51.5	(23)	
		MCWB	37.5	39.7	44.3	49.8	54.2	58.9	60.9	60.1	56.0	51.3	44.7	39.9	(24)	
	10%	DB	44.2	48.1	59.1	66.3	76.5	87.6	93.6	93.1	85.6	73.2	56.1	47.9	(25)	
		MCWB	35.7	38.3	43.1	48.3	53.3	58.5	60.6	59.4	55.2	50.3	43.7	38.5	(26)	
	0.4%	WB	42.3	45.6	49.4	54.9	60.2	65.0	68.2	68.5	60.9	55.9	50.4	45.8	(27)	
		MCDB	50.9	55.5	63.4	69.9	77.8	87.3	94.0	92.4	85.3	75.5	59.9	56.6	(28)	
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.3	43.0	47.3	52.9	57.6	62.7	65.5	65.5	58.8	53.6	48.5	43.1	(29)	
		MCDB	47.7	51.9	63.6	68.5	76.3	86.6	90.8	89.9	84.3	73.0	58.5	52.6	(30)	
	5%	WB	38.3	40.6	45.5	51.3	56.0	60.9	63.6	62.3	57.3	52.3	46.6	41.1	(31)	
		MCDB	46.2	49.4	60.2	65.8	74.7	85.4	89.7	88.5	84.1	71.8	56.2	48.5	(32)	
	10%	WB	35.9	38.8	43.9	49.7	54.6	59.5	62.0	60.5	55.8	51.1	45.0	39.1	(33)	
		MCDB	42.9	47.1	57.0	63.0	72.6	84.4	89.1	89.3	82.5	69.9	54.1	45.8	(34)	
	5% DB	MDBR	20.2	20.2	22.8	23.3	27.6	32.6	32.9	34.5	35.5	29.5	21.6	19.8	(35)	
		MCDBR	24.1	25.6	28.3	28.8	32.6	35.7	35.7	36.4	37.5	34.2	27.6	26.2	(36)	
		MCWBR	14.2	14.2	13.5	12.5	13.3	14.9	13.9	14.1	15.4	15.6	14.2	15.0	(37)	
		MCWBR	21.5	22.1	24.3	25.2	29.6	33.5	32.7	34.1	35.2	29.9	22.7	21.8	(38)	
	5% WB	MCWBR	13.2	13.0	12.6	11.5	12.7	14.8	13.9	14.3	15.3	14.2	12.6	13.4	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.294	0.332	0.385	0.463	0.458	0.433	0.473	0.428	0.365	0.380	0.327	0.294	(40)	
	taud		2.370	2.257	2.098	1.915	1.957	2.032	1.929	2.053	2.225	2.151	2.321	2.402	(41)	
	Ebn at Noon		289	287	279	262	265	270	259	269	282	265	272	282	(42)	
	Edh at Noon		30	37	48	61	59	55	60	52	42	41	31	27	(43)	
(44) All-Sky Solar Radiation	RadAvg		926	1233	1532	1814	2195	2599	2452	2294	1975	1425	983	800	(44)	
	RadStd		75	113	133	154	159	114	92	78	71	99	98	94	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	+1.91	-2.36	-2.72	N/A	N/A	N/A	N/A	
(47)	Regional (6 neighbors)	+0.65	N/A	N/A	+1.31	-1.49	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page